NORMALITY DESCENDS AND ASCENDS ALONG ETALE HOMOMORPHISMS OF NOETHERIAN LOCAL RINGS

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file etale.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Source reference. Original reference 033G: sources/stacks-chow/tags.txt, source lines 4055–4055. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 039L: sources/stacks-chow/tags.txt, source lines 4276–4276. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 033C: sources/stacks-chow/tags.txt, source lines 4051–4051. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Proposition 0.1. *Let A, B be Noetherian local rings. Let $f : A \to B$ be an étale homomorphism of local rings. Then A is a normal domain if and only if B is so.*

Proof. See Algebra, Lemma 033G for descending normality. Conversely, assume A is normal. By assumption B is a localization of a finite type A-algebra B' at some prime $\mathfrak{q}$. After replacing B' by a localization we may assume that B' is étale over A, see Lemma 039L. Then we see that Algebra, Lemma 033C applies to $A \to B'$ and we conclude that B' is normal. Hence B is a normal domain. □

Date: stacks-project:etale.tex:local-snapshot-20261012; GNU FDL 1.2 or later.